

CCES – College of Computer Engineering and Sciences

Prince Sattam Bin Abdulaziz University

CS3401 – Algorithm Design and Analysis

Chapter 1

Introduction to Algorithm Design

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Chapter Overview

Why Algorithms Matter

What is an Algorithm?

Case Study — Greatest Common Divisor

Algorithm Correctness

Algorithm Efficiency

Algorithm Design

Section 1

Why Algorithms Matter

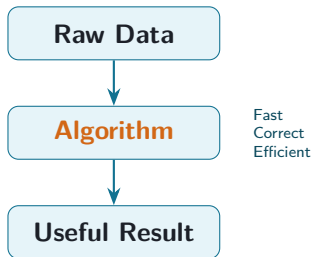
The Invisible Engine Behind Modern Technology

Every second:

- ▶ **Google** processes >100 000 search queries
- ▶ **Spotify** streams to >600 million users
- ▶ **GPS** recalculates routes for millions of drivers
- ▶ **Banks** clear thousands of transactions

The Question

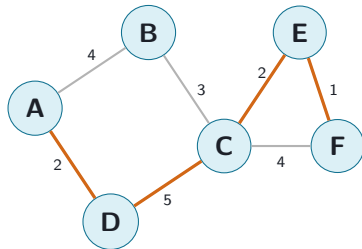
What tells a computer *how* to solve these problems correctly and at speed?



A Motivating Example: Route Planning

Problem: Given a map with n locations and road distances, find the *shortest path* from A to B.

- ▶ With 10 cities there are $\approx 181\,000$ possible routes
- ▶ With 20 cities: $\approx 6 \times 10^{16}$ routes
- ▶ Checking them all is **impossible**



Shortest: A→D→C→E→F (cost 10)

Algorithm to the Rescue

Dijkstra's algorithm finds the shortest path among millions of nodes in *milliseconds*.

Section 2

What is an Algorithm?

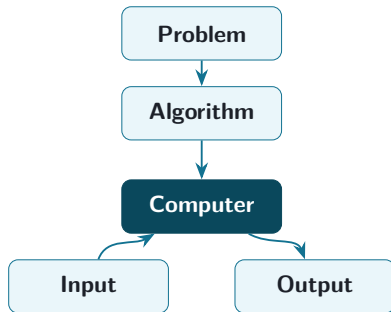
Definition

Algorithm

A **finite**, **unambiguous** sequence of instructions that, given some *input*, produces a corresponding *output* and terminates in a finite number of steps.

Key properties (DIOFEG)

- ▶ **D**efiniteness – each step is precisely defined
- ▶ **I**ntput – zero or more inputs
- ▶ **O**utput – at least one output
- ▶ **F**initeness – must terminate
- ▶ **E**ffectiveness – steps are basic and doable
- ▶ **G**enerality – works for a class of inputs



Special Features of Algorithms

- ▶ **Non-ambiguity** — every instruction has exactly one interpretation
- ▶ **Range of inputs** — must handle all valid inputs, not just examples
- ▶ **Multiple representations** — the same algorithm can be expressed as pseudocode, flowchart, or code
- ▶ **Multiple algorithms** — one problem can have many solutions
- ▶ **Different efficiencies** — different algorithms for the same problem can differ enormously in speed

Same Problem, Three Algorithms

Computing $\text{gcd}(60, 24)$:

1. **Euclid's algorithm** — elegant, fast
2. **Consecutive integer checking** — slow, limited
3. **Middle-school factoring** — complex, very slow

All three give the answer **12** — but not equally fast.

Section 3

Case Study — Greatest Common Divisor

Problem Statement

Greatest Common Divisor

Given two non-negative integers m and n (not both zero), find $\text{gcd}(m, n)$ — the *largest integer that divides both*.

Examples

- ▶ $\text{gcd}(60, 24) = 12$
- ▶ $\text{gcd}(60, 0) = 60$
- ▶ $\text{gcd}(7, 5) = 1$ (coprime)
- ▶ $\text{gcd}(0, 0) = ?$ **undefined**

Three classical approaches

Algorithm	Efficiency	Notes
Euclid's	$\mathcal{O}(\log \min(m, n))$	Best choice
Consecutive check	$\mathcal{O}(n)$	Fails for $n = 0$
Middle-school	Very slow	Ambiguous steps

Algorithm 1: Euclid's Algorithm

Key insight:

$$\gcd(m, n) = \gcd(n, m \bmod n)$$

Repeat until the second argument becomes 0.

Pseudocode

Input: $m, n \geq 0$, not both zero

Output: $\gcd(m, n)$

```
1: while  $n \neq 0$ 
2:    $r \leftarrow m \bmod n$ 
3:    $m \leftarrow n; n \leftarrow r$ 
4: end while
5: return  $m$ 
```

Trace: $\gcd(60, 24)$

m	n	$r = m \bmod n$
60	24	12
24	12	0
12	0	—

Result: $\gcd(60, 24) = 12$

Efficiency

The number of iterations is $\mathcal{O}(\log \min(m, n))$

Algorithm 1: Implementation

Recursive version

```
int gcd(int m, int n) {  
    if (n == 0) return m;  
    return gcd(n, m % n);  
}
```

Iterative version

```
int gcd(int m, int n) {  
    while (n != 0) {  
        int r = m % n;  
        m = n;  
        n = r;  
    }  
    return m;  
}
```

Why the iterative version is preferred

- ▶ No function-call overhead
- ▶ Constant stack space
- ▶ Same time complexity $\mathcal{O}(\log n)$

Convergence guarantee

After two consecutive iterations, the second argument decreases by *at least half* — hence $\mathcal{O}(\log n)$.

Algorithm 2: Consecutive Integer Checking

Idea: Start from $t = \min(m, n)$ and test each candidate downward until we find one that divides both.

Pseudocode

```
1:  $t \leftarrow \min(m, n)$ 
2: while  $t > 0$ 
3:   if  $m \bmod t = 0$  and  $n \bmod t = 0$  then
4:     return  $t$ 
5:   end if
6:    $t \leftarrow t - 1$ 
7: end while
```

Limitations

- ▶ **Fails** when either input is **zero**
($\min(m, 0) = 0$, loop never runs)
- ▶ Worst case: $\mathcal{O}(\min(m, n))$ iterations
- ▶ Much slower than Euclid's for large inputs

Example: $\text{gcd}(60, 24)$

- ▶ $t = 24$: $24 \nmid 60 \times$
- ▶ $t = 23 \dots 13$: $\times$ (11 wasted steps)
- ▶ $t = 12$: $12 \mid 60$ and $12 \mid 24 \checkmark$

Algorithm 3: Middle-School Procedure

Steps:

1. Find the prime factorization of m
2. Find the prime factorization of n
3. Identify all common prime factors (with multiplicity)
4. Multiply them to get $\text{gcd}(m, n)$

Example: $\text{gcd}(60, 24)$

$$60 = 2 \cdot 2 \cdot 3 \cdot 5 \quad 24 = 2 \cdot 2 \cdot 3 \cdot 2$$

$$\text{gcd}(60, 24) = 2 \cdot 2 \cdot 3 = 12$$

Why this algorithm is problematic

- ▶ Prime factorization is **not defined unambiguously** in the problem — it is itself a hard sub-problem
- ▶ Step 3 ("find common factors") is also not clearly specified
- ▶ Fails when an input equals 1
- ▶ Overall complexity: much worse than $\mathcal{O}(\log n)$

Lesson

Even a familiar school method can be a *bad* algorithm.

Sieve of Eratosthenes

Goal: Produce all primes $\leq n$ (needed by the middle-school method).

Pseudocode

```
1: for  $p \leftarrow 2$  to  $n$   $A[p] \leftarrow p$ 
2: end for
3: for  $p \leftarrow 2$  to  $\lfloor \sqrt{n} \rfloor$ 
4:   if  $A[p] \neq 0$  then
5:      $j \leftarrow p^2$ 
6:     while  $j \leq n$ 
7:        $A[j] \leftarrow 0$ ;  $j \leftarrow j + p$ 
8:     end while
9:   end if
10: end for
11: Output all  $A[p] \neq 0$ 
```

Trace for $n = 25$:

2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17
18	19	20	21	22	23	24	25

Primes ≤ 25 : **2, 3, 5, 7, 11, 13, 17, 19, 23**

Complexity: $\mathcal{O}(n \log \log n)$



Section 4

Algorithm Correctness

What Does "Correct" Mean?

Definition: Correctness

An algorithm is **correct** if, for *every* valid input, it terminates and produces the *right output*.

Why it is non-trivial:

- ▶ The set of valid inputs is usually *infinite*
- ▶ Testing a few examples is **not a proof**
- ▶ Intuition can be deceived by edge cases
- ▶ Formal proof (mathematical induction, loop invariants) is required

Common pitfall

"It works on the examples I tried" $\neq$ correct algorithm.

A single *counter-example* is enough to *disprove* correctness.

Good practice

After designing an algorithm, actively try to find inputs where it *fails*.

Case Study: The Shortest Tour Problem

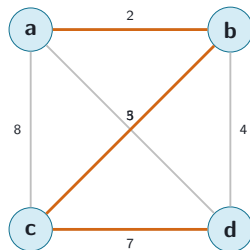
Problem: Given n cities with known pairwise distances, find the *shortest* round tour that visits each city exactly once (Travelling Salesman Problem — TSP).

Applications:

- ▶ Delivery route optimization
- ▶ Circuit board drilling
- ▶ DNA sequencing

Proposed heuristic: Nearest Neighbor (NN)

1. Start at any city
2. Repeatedly visit the *nearest unvisited* city
3. Return to the start

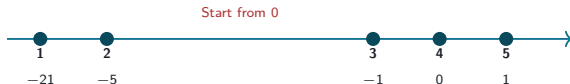


NN tour: $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a = 22$

Optimal: $a \rightarrow b \rightarrow d \rightarrow c \rightarrow a = 17$

Nearest-Neighbor Counter-Example

Points on a number line:



NN from point 4 (value 0):

- ▶ Visit 5 (dist 1), then 3 (dist 2), then 2 (dist 4), then 1 (dist 16), back to 4 (dist 22)
- ▶ Total: **45 units**

Optimal tour: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 1 = 26 \text{ units}$

Conclusion

The Nearest-Neighbor heuristic **does not always** produce an optimal tour.

Key lesson

- ▶ Seeking *counter-examples* is a vital part of algorithm design
- ▶ A plausible algorithm is not necessarily a correct one
- ▶ TSP has no known efficient exact algorithm (it is NP-hard)

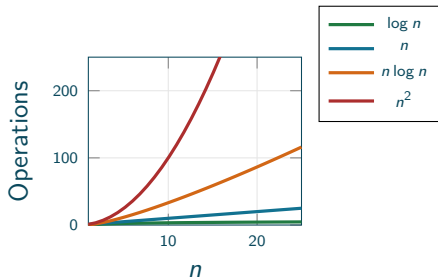


Section 5

Algorithm Efficiency

Why Efficiency Matters

- ▶ Hardware improvements are *incremental*; algorithmic improvements can be *exponential*
- ▶ A 10× faster CPU helps — switching from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$ helps *much more*
- ▶ Real systems must scale: from 10 000 to 10 000 000 users



Software always outstrips hardware

Even the fastest computer cannot save a poorly designed exponential-time algorithm for large inputs.

How to Measure Efficiency

Machine-independent approach:

- ▶ Analyse *pseudocode*, not actual code
- ▶ Assume an *idealized model*: each basic operation takes one unit of time
- ▶ Count the *number of basic operations* as a function of the *input size* n

$$T(n) \approx c_{\text{op}} \cdot C(n)$$

c_{op} : time per operation $C(n)$: operation count

Typical Input Size Measures

Problem	Size
Sorting	$n = \#$ elements
Matrix multiply	$n \times n$ dimension
Graph problems	$\#$ vertices + edges
Primality test	$\#$ digits of n

Typical Basic Operations

key comparison, addition,
multiplication, array access

Faster Algorithm vs. Faster CPU

Algorithm S1 (slow, on a fast machine):

- ▶ $T(n) = 2n^2$ operations, CPU @ 10^{10} ops/sec
- ▶ For $n = 10^6$: 2×10^{12} ops $\rightarrow$ **200 seconds**

Algorithm S2 (fast, on a slow machine):

- ▶ $T(n) = 50n \log_2 n$ ops, CPU @ 10^8 ops/sec
- ▶ For $n = 10^6$: $\approx 10^9$ ops $\rightarrow$ **10 seconds**

Takeaway

A better algorithm on a slower machine beats a worse algorithm on a faster machine for large inputs.

Algorithm	$n = 10^4$	$n = 10^6$
$\mathcal{O}(n^2)$	1.0 s	2.8 hr
$\mathcal{O}(n \log n)$	0.001 s	0.10 s
$\mathcal{O}(n)$	0.0001s	0.01 s

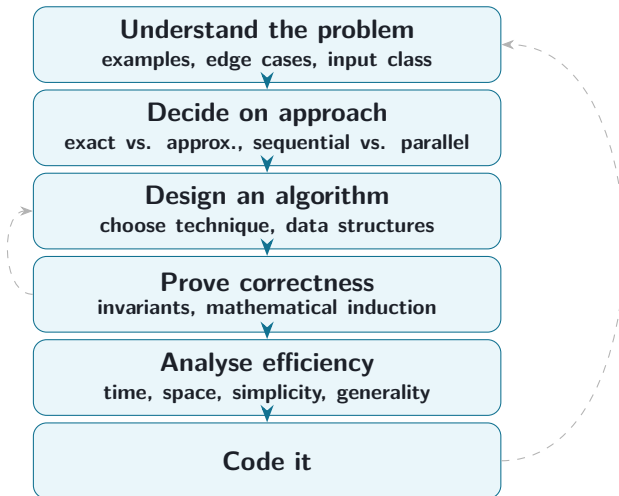
(Assuming 10^8 ops/sec)



Section 6

Algorithm Design

The Algorithm Design and Analysis Process



Algorithm Design Techniques

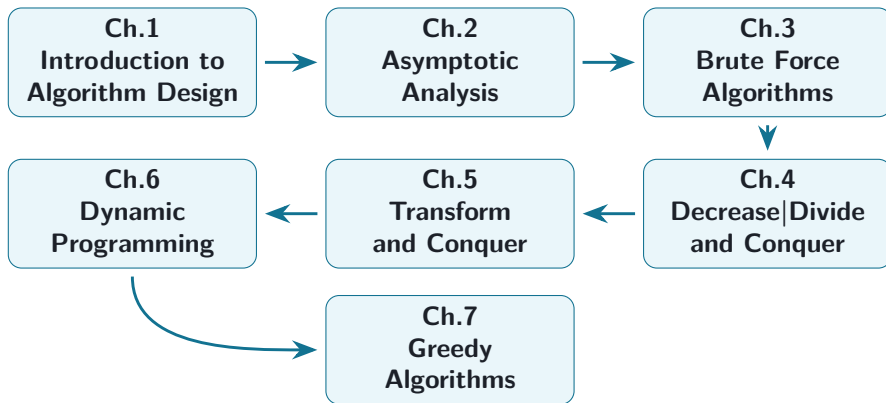
Techniques covered in this course

Brute Force	try all possibilities
Decrease & Conquer	reduce to smaller instance
Divide & Conquer	split into equal halves
Transform & Conquer	change representation
Dynamic Programming	overlapping sub-problems
Greedy Algorithms	locally optimal choices

Important Problem Types

Sorting	reorder a list
Searching	find an element
String processing	text, patterns
Graph problems	paths, flows, trees
Combinatorial	permutations, subsets
Geometric	points, lines, hulls
Numerical	equations, integrals

Course Roadmap



Chapter 1 Summary

Key concepts:

- ▶ An **algorithm** is a finite, unambiguous, effective sequence of steps producing a correct output
- ▶ The same problem can have *many* algorithms with very different properties
- ▶ **Correctness** requires proof — counter-examples are your best diagnostic tool
- ▶ **Efficiency** is measured in a machine-independent way using asymptotic notation

Three take-aways

1. Choose algorithms — don't just code the first idea
2. A faster algorithm on a slower machine beats a slower algorithm on a faster machine

Next chapter

Chapter 2: Asymptotic analysis, Big-O, Big- Θ , Big- Ω , recurrences, and running-time calculation